

Below is a **clear, endtoend instruction set** to build a digitaltwin matcher that uses **only structured EHR** to predict which antidepressant a new patient will most likely respond to. It includes data prep, feature engineering, narrative generation, LLM prompting, similarity scoring, and aggregation to a perdrug recommendation.

0) What you are building

Given a **new MDD patient** (the “index patient”), you will:

1. Convert their **structured EHR** into a **standard narrative** (semistructured text).
 2. Compare that narrative to a **library of historical patients** who already tried antidepressants and have outcomes.
 3. Use an **LLM to score similarity** between the index and each candidate.
 4. Combine similarity scores with historical outcomes to estimate which **drug** is most promising for the index patient.
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1) Set up your environment

- Programming language: **Python 3.10+**
 - Recommended libraries: pandas, numpy, scikit-learn, dateutil, faiss (or sklearn.neighbors), any LLM SDK you use.
 - Data format: CSV/Parquet or FHIR export mapped to flat tables.
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2) Ingest the minimum EHR tables

Create (or map) these tables; one row per record unless noted:

1. **patients**: patient_id, birth_date, sex_at_birth, marital_status, insurance, (optional) race_ethnicity
2. **diagnoses**: patient_id, icd10_code, diagnosis_date
3. **procedures**: patient_id, code, proc_date (ECT/TMS/ketamine identifiers)
4. **medications**: patient_id, rxnorm_ingredient, dose, route, days_supply, start_date, end_date
5. **labs**: patient_id, test_name, value, unit, result_date
6. **encounters**: patient_id, encounter_type (inpatient/ED/outpatient), admit_date, discharge_date
7. **appointments**: patient_id, scheduled_date, status (kept/noshow/cancel)
8. **pros** (if available): patient_id, instrument (PHQ9), item (Q1–Q9), value, date

If you have a FHIR export, map `Condition`→`diagnoses`, `MedicationRequest/Dispense`→`medications`, `Observation`→`labs/PROs`, `Encounter`→`encounters`, `Procedure`→`somatic treatments`.

3) Build the cohort and the index date

3.1 Identify antidepressants you will model

- SSRIs: sertraline, escitalopram, fluoxetine, citalopram, paroxetine
- SNRIs: venlafaxine XR, duloxetine
- Others: bupropion XL, mirtazapine, vortioxetine

3.2 Define MDD patients and newuser index

- A patient is eligible if they have **ICD10 F32.x/F33.x**.
- **Index date (t_0)** = first **new** antidepressant order/dispense (no supply for that ingredient in the prior **180 days**).

3.3 Fix your time windows

- **Baseline window** for features: **−730 to +0 days** around t_0 .
 - **History** for complexity/utilization: **3 years** before t_0 .
 - **Outcome window** for historical patients: **8–12 weeks** after t_0 (to judge response).
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4) Create value sets (code lists) you will use

- **Psych**: MDD (F32/F33), anxiety (F41.x), PTSD (F43.1x), OCD (F42.x), dysthymia (F34.1), insomnia (F51.x/G47.0), suicidality (R45.851; attempt X71–X83/T14.91), SUD (F10–F19).
- **Medical**: diabetes (E11.x), hyperlipidemia (E78.x), thyroid (E03/E05), chronic pain/fibromyalgia/neuropathy.
- **Somatic Tx**: ECT/TMS/ketamine codes (CPT/HCPCS).
- **Med classes**: SSRIs, SNRIs, bupropion, mirtazapine, vortioxetine, benzodiazepines, hypnotics, NSAIDs.

Keep these lists in JSON so the pipeline is reproducible.

5) Engineer the features you need (all computed from structured data)

Implement each as a small function that takes a patient's rows and returns a value. Use the baseline and history windows exactly.

1. **Age** at t_0 = floor((t_0 - birth_date)/365.25).
2. **BMI** = last measured weight/height in baseline (optional if available).
3. **Psych comorbidity flags** (yes/no): any anxiety/PTSD/OCD/dysthymia code in the **last 24 months**.
4. **SUD flag**: any F10–F19 in last 24 months.
5. **Suicidality flag (12 months)**: ideation or attempt codes.
6. **Pain signal**: (a) presence of chronic pain diagnoses, (b) **NSAID burden** = number of distinct NSAID ingredients with ≥ 7 days' coverage in baseline (0/1/ ≥ 2).
7. **Metabolic/medical**: diabetes, hyperlipidemia, thyroid disorder (ever), and most recent **A1c/LDL/TSH** bands in the last 12 months.
8. **Polypharmacy** at baseline: count of distinct **active** medications with ≥ 14 days' overlap with t_0 ; cap at 20.
9. **Benzodiazepine days** in baseline (0 / 1–30 / >30).
10. **Psych utilization**: psychiatric **inpatient days** and **ED visits** in **12 months** and in **3 years**.
11. **Prior adequate antidepressant trials (24 months)**: for each class, did the patient have ≥ 6 weeks at a therapeutic dose? Return counts by class.
12. **PHQ9** (if available): total score and items Q1 (anhedonia), Q2 (mood), Q3 (sleep), Q4 (fatigue), Q9 (thoughts of death).
13. **Noshow ratio**: appointments with status=noshow / scheduled in the past 12 months.
14. **3year complexity score** (simple version): zscore sum of (Elixhauser count, total inpatient days, ED visits, medication class count). Rescale to 0–1.
15. **Mechanistic indices (0–1)** (optional but helpful):
 - Threat/Anxiety = normalized sum of anxiety/PTSD/SUD/suicidality + benzo days + ED crisis.
 - Metabolic/Somatic = normalized sum of BMI/A1c/LDL/thyroid + pain flag.
 - TreatmentBurden = prior adequate trials + psych admissions + augmentations.
 - Polypharmacy = minmax of activemed count.

Store all feature values in a single **row per patient** (wide table).

6) Generate the standard narrative (semistructured text)

Use **exact section headers** and **key: value** pairs so every patient looks the same. For missing values, write **Missing**; for negative facts, write **Absent**.

COHORT & INDEX

Condition: MDD | Index date: 2025-03-04 | Baseline window: -90..+7 days

DEMOGRAPHICS & VITALS

Age: 47 y | Sex: F | BMI: 31.8 kg/m² | BP: 132/82 mmHg

PSYCH HISTORY (STRUCTURED)

Anxiety: Present | PTSD: Present | OCD: Absent | Dysthymia: Absent
SUD (24m): Alcohol use disorder | Suicidality (12m): Absent | Insomnia:
Present

MEDICAL COMORBIDITY

Diabetes: Absent (A1c 5.6%) | Hyperlipidemia: Present (LDL 158 mg/dL)
Thyroid disorder: Absent (TSH 1.9) | Chronic pain: Lumbar radiculopathy

LABS (MOST RECENT IN 12m)

A1c: 5.6% | LDL: 158 | HDL: 44 | TG: 210 | TSH: 1.9 | eGFR: 86

TREATMENT EXPOSURE (24m HISTORY + BASELINE)

Prior adequate AD trials: SSRI=1, SNRI=0, Bupropion=1, Mirtazapine=0,
Vortioxetine=0
Benzodiazepine days (90d): 22 | Hypnotics: Absent | Augmentation: None
Somatic treatments: None | Psychotherapy visits (12m): 3

MEDICATION BURDEN

Active meds at baseline: 12 (Polypharmacy=High) | Strong CYP inhibitors: None
NSAID burden: 2 (ibuprofen, naproxen)

UTILIZATION

Psych inpatient days: 0 (12m) / 0 (3y) | ED psych visits: 1 (12m) / 2 (3y)

SOCIODEMOGRAPHICS / ACCESS

Marital: Separated/Divorced | Employment: Part-time | Insurance: Commercial
No-show ratio (12m): 0.22 | SDoH flags: None recorded

BASELINE SYMPTOMS (STRUCTURED PROs)

PHQ-9 total: 17 | Anhedonia: 2 | Depressed mood: 3 | Sleep: 3 | Fatigue: 3 |
Thoughts of death: 0

INDICES (0-1; SITE-SCALED)

Threat/Anxiety: 0.66 | Metabolic/Somatic: 0.58 | Treatment-Burden: 0.53 |
Polypharmacy: 0.78
3-year Complexity: 0.63

SAFETY / CONTRAINDICATIONS

QT risk: No | Seizure risk: No | Uncontrolled HTN: No

MISSINGNESS

hs-CRP: Missing | Family history: Missing

Also append a oneline “**machine band**” for deterministic parsing:

AGE=45-54;SEX=F;BMI=31.8;ANXIETY=Y;PTSD=Y;SUD=AUD;SUICIDALITY12M=N;NSAID_CT=2
;POLYPHARM_CT=12;A1C=5.6;LDL=158;TSH=1.9;PRIOR_AD=SSRI:1,BUP:1;PHQ9=17;NO_SHO
W=0.22;...

Generate this narrative for:

- Every **historical** patient (plus their **8–12 week outcomes**).
- Every **new** index patient (without outcomes).

7) Build the digitaltwin library

For each historical patient:

- Store (narrative_text, feature_row, drug_taken, response/remission, AE_discontinuation).
- Consider **one record per drug trial** if a patient had multiple trials (each has its own t_0 and outcome).

8) Fast prefilter before LLM (saves cost/time)

Compute a simple **numeric vector** for each patient (e.g., age, PHQ9 total, anxiety flag, SUD flag, polypharmacy count, NSAID count, A1c band, LDL band, prior trial counts, 3year complexity).

Use **cosine similarity** / ANN (FAISS) to select the top **K=200** historical candidates for the index patient. These are the only ones you'll send to the LLM.

9) Prompt the LLM to judge pairwise similarity

9.1 System prompt (set once)

You are a clinical similarity scorer for major depressive disorder (MDD). Compare two patient narratives that contain only structured EHR data from a fixed baseline window.

Judge similarity only on factors that affect antidepressant response/tolerability.

Treat "Missing" as unknown (neutral). Treat "Absent" as truly absent. Do not infer beyond text.

Weight these dimensions when scoring (sum=100):

- Baseline symptom phenotype (PHQ-9 subitems): 25
- Psychiatric comorbidity & threat/anxiety/trauma & SUD/suicidality: 20
- Medical/metabolic & pain/NSAIDs: 20
- Treatment exposure & medication burden (polypharmacy, prior adequate trials, 3y complexity): 20
- Social/functional access (no-show, marital/employment/SDoH): 10
- Safety flags (contraindications): 5

Return JSON only.

9.2 User prompt template (for each candidate)

INDEX_PATIENT:
<index patient narrative>

CANDIDATE_PATIENT:
<candidate narrative>

Return this JSON:

```
{
  "overall_similarity": 0-100,
  "phenotype": 0-100,
  "psych_comorbidity": 0-100,
  "metabolic_pain": 0-100,
```

```

"treatment_burden": 0-100,
"social_functional": 0-100,
"safety": 0-100,
"top_similarity_drivers": ["short phrase x5-8"],
"key_mismatches": ["short phrase x3-5"]
}

```

Scoring rules:

- Same band (e.g., NSAID 0 vs 1) = small penalty; different band (0 vs ≥2) = large penalty.
- Reward matches on polypharmacy, pain/NSAIDs, anxiety/fatigue most.
- If a dimension has any "Missing", exclude that dimension and renormalize the weights.

Cache results so the same pair is not rescored twice.

10) Turn similarity scores into drug predictions

For the K candidates, group by the **drug they actually took**. For each drug (d):

1. Normalize the LLM **overall_similarity** to $[0, 1]$ (e.g., divide by 100).
2. Compute **similarityweighted response rate**:

$$\hat{p}\{\text{resp}\}(d) = \frac{\sum_{i \in d} s_i \cdot \mathbf{1}[\text{response}_i = 1]}{\sum_{i \in d} s_i}$$
3. Compute **similarityweighted AE discontinuation** the same way; optionally remission too.
4. Compute **Expected Net Benefit**:

$$\text{ENB}(d) = \hat{p}\{\text{resp}\}(d) - w \cdot \hat{p}\{\text{AE-DC}\}(d)$$

where (w) is a penalty (e.g., 0.5).
5. Pick the drug with the **highest ENB** (report top3 with 95% Wilson CIs using the weighted counts).

11) Present the result (clinicianfacing card)

- **Top recommendation:** Drug X — Response 0.58 (CI 0.49–0.66), AEDC 0.09 → ENB 0.53
- **Why:** Pull 3–5 bullets from `top_similarity_drivers` most frequently seen among twins for Drug X.
- **Alternatives:** 2nd/3rd drugs with short reasons.
- **Caveats:** list `key_mismatches` that could reduce confidence.

12) Evaluate your system

- **Face validity:** sample 50 matches; have a clinician rate agree/disagree.
- **Backtesting:** split historical data into train (twins) / test (queries). For each test index patient, see if the drug you would have recommended matches the one that actually worked better than average.
- **Ablation:** remove one feature family at a time (polypharmacy, NSAIDs, etc.) and confirm performance drops (sanity check).
- **Fairness:** ensure race/ethnicity (if present) is not used in scoring; compare performance across age/sex groups.

13) Practical guardrails

- **No PHI** in narratives (no names/addresses).
- **Missing data:** mark as “Missing”; never invent values.
- **Leakage control:** for index patients you’re predicting, do **not** include anything measured after t_0 .
- **Reproducibility:** version your code lists and prompts; log every narrative and LLM JSON returned.

14) Minimal artifacts you should deliver

1. `features.py` — functions to compute all features from raw tables.
2. `narrative_template.py` — renders the narrative + machine band.
3. `prefilter.py` — builds numeric vectors and retrieves topK candidates.
4. `similarity_llm.py` — sends the pairwise prompt and parses JSON.
5. `aggregate.py` — computes perdrug weighted outcomes and ENB.
6. `report.py` — generates a clinician card for the index patient.
7. `README.md` — onepage runbook with cohort definitions and windows.

15) Quick checklist (laminated card)

- Cohort (MDD, newuser index).
- Features computed in fixed windows.
- Narrative generated (sections + machine band).

- Candidate prefilter ($K \approx 200$).
- LLM similarity JSON for each pair.
- Similarityweighted drug outcomes + ENB.
- Top3 recommendation with drivers and caveats.
- Logs, audits, and evaluation.

If you want, I can also provide a small **starter CSV schema** and **template code** that reads 10 mock patients and outputs narratives and similarity scores so a student can run the entire pipeline in under an hour.

Below is a precise, implementation-oriented outline of what to do **after your digitaltwin matcher exists** to support:

1. **Prediction of TRD** (treatment-resistant depression risk), and
2. **Counterfactual treatment optimization** (expected outcomes for different *subsequent* antidepressants).

I am assuming your matcher already produces:

- A **standardized narrative** for a patient “state” (built from structured EHR only), and
- A function $SIM(A, B)$ that returns a **similarity score** (0–100) between two narratives, plus optional driver strings.

I am also assuming you have a historical EHR cohort with antidepressant exposures and outcomes.

A. Shared prerequisites (do these once)

A1) Define consistent time anchors and state snapshots

You must define **exactly** when predictions are made, and exactly what data are allowed.

Time anchors

- **t_0 (line start):** date of a new antidepressant start (order/dispense) for an MDD episode.
- **Decision points** (examples):
 - **$t_1 = t_0 + 8$ to 12 weeks** (end of acute trial window)
 - **$t_1' = \text{early stop}$** (if discontinuation/switch before 8 weeks)

State snapshot windows (must be fixed)

At each decision point t :

- **Baseline/features window:** $t - 90 \text{ days}$ to $t + 7 \text{ days}$ (structured EHR only)
- **Recent history window:** $t - 3 \text{ years}$ to t (utilization/comorbidity burden)

Leakage rule: For a prediction at time t , you cannot use any structured data recorded after $t + 7$ days.

A2) Build “treatment lines” and label adequacy

For every patient, segment EHR medication history into antidepressant **lines**.

A line record should have:

- `patient_id`
- `line_number` (1, 2, 3...)
- `line_start_date` (t_0)
- `drug` (ingredient), `class`, `dose`, `duration`, `days_supply`, `adherence` proxies (PDC/MPR)
- `adequate_trial` (Yes/No)

Adequate trial rule (example; choose and lock):

- `Duration` ≥ 6 weeks at or above therapeutic dose **AND**
- `PDC` ≥ 0.8 (or you can allow 0.7 with sensitivity analysis)

Store dose thresholds in a versioned lookup table.

A3) Define outcomes in an EHRimplementable way

You need two outcome families:

(i) TRD outcome (for part 1)

Define TRD over a horizon (commonly 12 months after line 1 start). A practical operational TRD label:

TRD = 1 if, within **12 months of line 1 start**, the patient has either:

- **≥2 adequate antidepressant trials** (different classes or distinct ingredients) with evidence of nonresponse, OR
- **Escalation** to ECT/TMS/esketamine/ketamine, OR
- Persistent high severity if you have structured symptom scales (optional rule)

Also store:

- TRD_date = earliest date TRD criteria are met (for timetoevent analyses)

(ii) Response/remission outcomes (for part 2)

For each line (or each switch event), define:

- **Acute response** at 8–12 weeks (preferred):
 - PHQ9 decrease ≥50% OR PHQ9 ≤9 (choose one)
- **Remission**:
 - PHQ9 ≤4
- **AE discontinuation**:
 - stop/switch within ≤28 days plus AEproxy codes or intolerability flags (structured proxies)

If no PHQ9 exists, define response proxies (switch/augmentation patterns), but keep them separate and report performance separately for “PRO available” vs “PRO missing.”

A4) Decide what a “digital twin set” is

Given a query state Q , you will retrieve a set of historical states $\{H_i\}$ with similarity weights $\{w_i\}$.

Recommended weighting function

Let $s_i = \text{SIM}(Q, H_i)$ in $[0, 100]$. Convert to weights:

- Option 1 (simple): $w_i = (s_i/100)^p$ with $p \in [2, 6]$
- Option 2 (softmax): $w_i = \exp(s_i/\tau) / \sum \exp(s_j/\tau)$ (τ controls sharpness)

Effective sample size (ESS) check

Compute:

$$ESS = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2}$$

Use ESS to decide whether the twin set is reliable. Example:

- $ESS \geq 50$: high confidence

- ESS 20–49: moderate
- ESS < 20: low confidence (report “insufficient similar patients”)

A5) Fix evaluation splits before you start

To measure “how well it works,” you need a **heldout** evaluation.

Use one of:

- **Time split** (recommended): train/library built from earlier years, test on later years
 - **Site split** (if multisite): build twins from sites A/B, test on C
 - **Patient split** (minimum): standard train/test split by patient
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B. (1) TRD prediction using digital twins

B1) Create the TRD prediction dataset (line1 anchored)

For every historical patient meeting inclusion criteria:

1. Identify their **line 1 start** t_0 .
2. Build the **state snapshot narrative at t_0** (structured EHR only, baseline/history windows).
3. Assign the TRD label:
 - TRD_12m (0/1)
 - optionally TRD_date for timetoevent

Store as rows:

- state_id, patient_id, t0, narrative_text, machine_band, TRD_12m, TRD_date

This table is your TRD twin library.

B2) TRD prediction for a new patient Q

Given a new patient starting an antidepressant at t_0 (Q):

Step 1 — Build Q’s t_0 narrative

Generate the standardized narrative (diagnoses, labs, meds, procedures, sociodemographics, utilization) using the **exact same template** as your library.

Step 2 — Prefilter candidates (fast, nonLLM)

Before calling the LLM pairwise scorer:

- Use a numeric vector (age band, polypharmacy band, comorbidity burden, anxiety flag, pain/NSAIDs, metabolic bands, prior antidepressant exposure)
- Retrieve top $K_{pre} = 200-1000$ candidates via approximate nearest neighbors.

Step 3 — LLM similarity scoring

For each prefiltered candidate H_i , compute:

- $s_i = \text{SIM}(Q, H_i)$
Keep top $K = 100-300$ by s_i .

Step 4 — Compute TRD risk estimate

Compute similarity weights w_i and estimate:

$$\hat{p}_{TRD}(Q) = \frac{\sum_i w_i \cdot TRD12m(H_i)}{\sum_i w_i}$$

Also compute uncertainty:

- Bootstrap the twin set (weighted bootstrap) to get a 95% interval, **or**
- Use an approximate effective binomial with ESS.

Step 5 — Produce calibrated risk tiers (optional but recommended)

Map predicted probability to a tier:

- Low < 0.20
 - Moderate $0.20\text{--}0.50$
 - High > 0.50
- (Thresholds should be chosen on the training set to optimize clinical utility.)

Step 6 — Report confidence diagnostics

Always report:

- ESS
- max similarity score and distribution (e.g., median s)
- missingness flags (e.g., “no PHQ9 available” if relevant)

B3) How to evaluate TRD prediction performance (precisely)

On the heldout test set:

Step 1 — Run the full pipeline

For each test patient Q :

- build narrative at t_0
- retrieve twins from the **training library only**
- compute $\hat{p}_{TRD}(Q)$

Step 2 — Compute discrimination

- ROCAUC
- PRAUC (important if TRD prevalence is low)

Step 3 — Compute calibration

- Calibration intercept/slope
- Reliability curve (bin predictions into deciles)
- Brier score

Step 4 — Clinical utility

- Decision curve analysis (net benefit vs treatall/treatnone)
- Sensitivity/specificity at chosen thresholds

Step 5 — Subgroup and robustness checks

Stratify performance by:

- age bands
- sex
- sites
- baseline polypharmacy bands
- PRO availability vs missing

Step 6 — Ablations to prove the “twin” method adds value

Rerun with:

- LLM similarity replaced by numeric cosine similarity only
 - No polypharmacy variables
 - No anxiety/pain variables
- Performance drop supports that the twin approach is using meaningful signals.

C. (2) Counterfactual treatment optimization for subsequent antidepressants

This is the more complex part because you are estimating outcomes under **different next treatments** from a given patient state.

C1) Define the decision point and “state at switch”

Pick the scenario you want to optimize.

Most common: **after nonresponse or intolerance to line 1**, decide line 2.

Define a switch decision point t_{switch} :

- Either at 8–12 weeks if nonresponse, OR
- at earlier discontinuation

A **switchstate** must include structured information about:

- which drug was tried (line 1), dose adequacy, duration, adherence proxy
- whether response occurred (PHQ9 change if available, otherwise proxy)
- whether adverse effects occurred (structured proxies)
- updated comorbidity/utilization within the last 90 days

You will generate a standardized “state narrative at t_{switch} .”

C2) Build the historical “switch event” library

Create one row per observed decision to start a new antidepressant after line 1.

For each historical patient:

1. Identify line 1 start t_0 .
2. Identify **line 2 start** t_2 (or augmentation if you include it as an option).
3. Define $t_{\text{switch}} = t_2$ (the time when the decision was made).
4. Generate the **switchstate narrative** at t_{switch} , using only data up to $t_{\text{switch}} + 7d$.

Attach:

- `chosen_next_drug = drug(line 2)`
- Outcomes after that choice:

- response_8to12w on line 2
- remission_8to12w
- AE_discontinue_0to4w
- TRD_12m_from_t0 (optional secondary)
- time_to_next_switch (proxy of ineffectiveness/tolerability)

This is your line2 twin library.

If you want multistep optimization (line 3, line 4), repeat the same procedure to build libraries for each line.

C3) Counterfactual prediction for a new patient at switch time

Given a query patient Q at $t_{\text{switch}}(Q)$:

Step 1 — Build Q 's switchstate narrative

This must include:

- line 1 drug/class, adequacy, adherence proxy
- response status (structured)
- any intolerance flags (structured)
- updated comorbidities/utilization and current labs/med burden

Step 2 — Enumerate feasible candidate next treatments

Create a candidate set $D(Q)$ by applying rulebased safety/feasibility gates:

- avoid drugs already failed adequately (unless partial response + AE, where withinclass retry might be reasonable)
- contraindications (seizure risk for bupropion; QT rules; BP rules)
- DDI burden constraints if polypharmacy is high

This step is essential to avoid recommending implausible counterfactuals.

Step 3 — For each candidate drug d , retrieve drugspecific twins

For each $d \in D(Q)$:

1. Prefilter within the historical switch library to rows where `chosen_next_drug == d`.
2. ANN prefilter (numeric) to top K_{pre} candidates.
3. Compute LLM similarity scores $s_i(d)$ between Q and each candidate.
4. Keep top $K(d)$ candidates and compute weights $w_i(d)$.

Step 4 — Estimate counterfactual outcomes for each drug

For each drug d , estimate:

- Expected response:

$$p_{\text{resp}}^{\wedge}(d|Q) = \frac{\sum_i w_i(d) \cdot \text{resp}_i}{\sum_i w_i(d)}$$

- Expected AE discontinuation:

$$[\hat{p}_{AE}(d|Q) = \frac{\sum_i w_i(d) \cdot AE_i}{\sum_i w_i(d)}]$$

- Optional: expected timetonextswitch (weighted median survival)

Step 5 — Compute a utility and choose the optimal treatment

Define an explicit utility function (must be fixed before evaluation), e.g.:

$$[U(d|Q) = \hat{p}_{resp}(d|Q) - \lambda \hat{p}_{AE}(d|Q) - \gamma \cdot Complexity(d)]$$

Where:

- $Complexity(d)$ could be a simple 0/1 indicator for drugs requiring more monitoring
- λ and γ are chosen by clinical leadership (and can be tuned in sensitivity analyses)

Choose:

$$[d^* = \arg \max_{d \in D(Q)} U(d|Q)]$$

Step 6 — Report uncertainty and overlap

For each d , report:

- $ESS(d)$
- 95% intervals for response and AE estimates (bootstrap)
- “Not enough similar patients for d ” if $ESS(d)$ too low

C4) How to evaluate counterfactual treatment optimization (precisely)

This is where you determine “how well it works.” Because counterfactuals aren’t directly observed, you need multiple evaluation layers.

Layer 1 — Outcome prediction accuracy on the *observed* treatment

On a heldout set of switch events $\{Q_j\}$:

1. For each Q_j , run the pipeline and compute predicted outcomes **for the drug they actually received**, d_{obs} .
2. Compare:
 - predicted $\hat{p}_{resp}(d_{obs}|Q_j)$ vs actual $resp_j$
 - predicted $\hat{p}_{AE}(d_{obs}|Q_j)$ vs actual AE_j

Compute:

- Brier score / log loss for response
- Calibration curves (predicted vs observed)
- AUROC for response as a probability score

This tests whether your twin method can predict outcomes *when the treatment is known*.

Layer 2 — Policy evaluation (does choosing d^* improve outcomes?)

You need a way to estimate the value of the “choose d^* ” policy compared to standard care.

Two practical approaches:

(A) Agreement subset evaluation (simple, conservative)

1. Identify cases in test data where clinician choice matched the policy:
$$d_{obs} == d^*(Q_j)$$
2. Compare observed outcomes in the agreement group vs nonagreement group, adjusting for baseline differences (at minimum stratify on severity proxies and comorbidity burden).

Limitation: selection bias (agreement is not random), but it provides a sanity check.

(B) Offpolicy evaluation (recommended)

You treat the clinician’s choice as a behavior policy and estimate the value of your policy.

Minimum viable option: **selfnormalized inverse propensity weighting (SNIPS)**.

Steps:

1. Fit a propensity model on the training set to estimate:
$$\pi(d|x) = P(\text{clinician chose } d \mid \text{state features } x)$$

(Use the numeric features that also drive the narrative; do not use future outcomes.)
2. For each test case j , define indicator:
$$I_j = 1 \left[d_{obs} == d^*(Q_j) \right]$$
3. Estimate policy value:
$$V_{SNIPS}^{\wedge} = \frac{\sum_j I_j \cdot \frac{resp_j}{\pi(d_{obs}|x_j)}}{\sum_j I_j \cdot \frac{1}{\pi(d_{obs}|x_j)}}$$
4. Compare to baseline value under observed clinician policy (the raw mean response) or to a reference policy (e.g., “always SSRI”).

Best practice: also compute a **doubly robust (DR)** estimate (propensity + outcome model), but SNIPS is often enough to begin.

Report confidence intervals via bootstrap.

Layer 3 — Sensitivity analyses (required to trust results)

You should rerun evaluations under variations to check stability:

- Vary K and weighting sharpness (p or τ)
- Restrict to cases with high overlap (ESS thresholds)
- Separate “nonresponse” vs “intolerance” switch reasons (structured proxy)
- Evaluate per drug class (SSRI vs SNRI vs bupropion vs mirtazapine vs vortioxetine)

- Check performance when PROs exist vs not

If the method only “works” in one narrow configuration, that is informative.

Layer 4 — Negative control / leakage checks

To ensure the method isn’t cheating:

- Confirm that predictions do not change when you remove postdecision data.
 - Evaluate a negativecontrol outcome that shouldn’t differ by antidepressant choice (sitespecific administrative outcome) to detect confounding artifacts.
-

D. Practical deliverables to make performance auditable

To “precisely determine how well the digital twin approach works,” you should generate and version the following artifacts:

D1) A reproducible evaluation report per run

For TRD and for line2 optimization:

- dataset definition + inclusion/exclusion
- code/value set versions
- narrative template version and prompt version
- K, weighting method, ESS thresholds
- metrics: AUC, calibration, Brier, SNIPS/DR policy value
- subgroup analyses
- failure analysis on low ESS cases

D2) Perpatient audit logs (for debugging)

For each query:

- top 20 twins with similarity scores
- computed weights and ESS
- predicted outcomes and intervals
- the LLM’s “drivers” and “mismatches” text

This is crucial for diagnosing why the model behaves as it does.

E. Minimal “cookbook” summary

(1) TRD Prediction

1. Build line1 twin library: narrative at t_0 + TRD label (12m).
2. For a new t_0 patient: retrieve top twins → weight them → compute weighted TRD probability + ESS + CI.
3. Evaluate on heldout: AUC, calibration, Brier, decision curves, ablations.

(2) Counterfactual NextTreatment Optimization

1. Build switchevent library: narrative at t_{switch} + chosen next drug + outcomes after switch.
2. For a new switch patient: for each candidate drug d , retrieve twins restricted to d → weight → estimate response/AE → compute utility → choose argmax.

3. Evaluate:

- prediction accuracy for the observed drug (Brier/calibration),
- policy value via SNIPS/DR,
- robustness via ESS thresholds and sensitivity analyses.

If you want me to align these steps even more tightly to the exact mechanistic splits and falsification logic from your previously attached tree document, please reupload that document—some previously uploaded files have expired in my workspace, so I cannot reopen them on demand without a reupload.